

Iteration 3 - Crispin Huan

Hand off from Kolade (I2)

$$w_2 = [-1.33331, 1.1765] \quad b_2 = [1.0185, 0.0121]$$

Step 1: Predict $y_i = w_2 x_i + b_i$

$$y_1 = (-1.33331)(1) + (1.1765)(3) + 1.0185 = 3.2115$$

$$y_2 = (-1.33331)(1) + (1.1765)(10) + 0.0121 = 7.3447$$

Step 2: Error = $y_i' - y_i$

$$\text{Error}_1 = 3.2115 - 5 = -1.7851$$

$$\text{Error}_2 = 7.3447 - 6 = 1.3447$$

Step 3: Cost $J(w, b) = \frac{1}{n} \sum_{i=1}^n (y_i' - y_i)^2, n=2$

$$J = \frac{1}{2} [(3.2115 - 5)^2 + (7.3447 - 6)^2]$$

$$J = \frac{1}{2} [(-1.7851)^2 + (1.3447)^2] = 2.4574$$

Step 4 (Chain rule): $dJ/dw = dJ/dy' \cdot dy'/dw$

$$dJ/dy' = (2/n) \cdot \text{Error} \quad \leftarrow \text{order derivative}$$

$$dy'/dw = X \quad \rightarrow dJ/dw = (2/n) \cdot X^T \cdot \text{Error}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 10 & 1 \end{bmatrix}$$

$$dJ/dw_1 = (1) (-1.7851) + (1) (1.3447) = -3.5937$$

$$dJ/dw_2 = (3) (-1.7851) + (10) (1.3447) = 8.0917$$

Iteration 3 (cont'd)

Step 5 (Chain rule): $dJ/db_i = dJ/dy' \cdot dy'/db_i$

$$dy'/db_i = 1 \quad (\text{since } y_i' = \dots + b_i)$$

$$\Rightarrow dJ/db = (2/n) \cdot \text{Error} \quad (n=2, \text{ so } 2/n=1)$$

$$dJ/db_1 = -1.7851 \quad dJ/db_2 = 1.3447$$

Step 6: Update (learning rate $a = 0.01$)

$$w_{3,1} = -1.33331 - (0.01)(3.5937) = -1.3650$$

$$w_{3,2} = 1.1765 - (0.01)(8.0917) = 1.0956$$

$$b_{3,1} = 1.0185 - (0.01)(-1.7851) = 1.0364$$

$$b_{3,2} = 0.0121 - (0.01)(1.3447) = 0.8987$$

Hand off to abdabji (Iteration 4)

$$w_3 = [-1.3650, 1.0956] \quad b_3 = [1.0364, 0.8987]$$

Trend: Acc fell from previous iterations towards 2.1934, m and b shift in small, consistent steps each update. Confirms gradient is descent is reducing error, not diverging.